

# Online Bookstore Management & Analytics

## Project Documentation

*Comprehensive Data Analysis & Business Intelligence Report*

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# 1. Executive Summary

This document presents the comprehensive project documentation for the Online Bookstore Mangement & Analytics system — a data-driven retail intelligence platform covering book inventory, customer profiles, sales transactions, return processing, and customer review analysis. The datasets collectively span 25 books, 60 customers, 150 orders, 9 returns, and 200 reviews, providing a rich foundation for business intelligence and decision-making.

## 1.1 Key Performance Indicators

|                                |                                       |                            |                                      |
|--------------------------------|---------------------------------------|----------------------------|--------------------------------------|
| <b>25</b><br>Books in Catalog  | <b>60</b><br>Registered Customers     | <b>150</b><br>Total Orders | <b>INR 1,76,689</b><br>Total Revenue |
| <b>200</b><br>Customer Reviews | <b>3.78 / 5</b><br>Avg. Review Rating | <b>6.0%</b><br>Return Rate | <b>62%</b><br>Delivery Success Rate  |

## 1.2 Project Objectives

- Consolidate multi-source retail data into a unified, query able structure
- Identify top-performing books, categories, and revenue trends
- Profile customer demographics, purchase behaviour, and preferences
- Analyse order fulfilment, payment methods, and coupon effectiveness
- Examine return patterns to reduce losses and improve service quality
- Leverage customer reviews for product quality insights and sentiment analysis

# 2. Dataset Overview

The project integrates five interrelated CSV datasets. Each dataset represents a distinct business domain and is linked via common identifiers such as Book\_Id and Customer\_Id.

## 2.1 Data Dictionary Summary

| Dataset   | File Name     | Records | Fields | Primary Key | Description                                   |
|-----------|---------------|---------|--------|-------------|-----------------------------------------------|
| Books     | Books.csv     | 25      | 15     | Book_Id     | Book catalog with pricing, ratings, and stock |
| Customers | customers.csv | 60      | 11     | Customer_Id | Customer profiles, demographics, membership   |

| Dataset | File Name     | Records | Fields | Primary Key | Description                                     |
|---------|---------------|---------|--------|-------------|-------------------------------------------------|
| Orders  | Orders.csv    | 150     | 10     | Order_Id    | Sales transactions with payment and status info |
| Returns | Returns.csv   | 9       | 16     | Return_Id   | Return and refund tracking per order            |
| Reviews | Reviews_1.csv | 200     | 11     | Review_Id   | Customer reviews and ratings per book           |

2.2 Entity Relationship

The datasets are linked as follows:

- Books (Book\_Id) <--> Orders (Book\_Id): Each order references a specific book
- Customers (Customer\_Id) <--> Orders (Customer\_Id): Each order is placed by a customer
- Orders (Order\_Id) <--> Returns (Order\_Id): Returns trace back to original orders
- Books (Book\_Id) <--> Reviews (Book\_Id): Reviews are tied to specific books
- Customers (Customer\_Id) <--> Reviews (Customer\_Id): Reviews are attributed to customers

3. Books Catalog Analysis

3.1 Catalog Overview

The books dataset contains 25 titles across 7 categories. All books are available in English, Hindi, or Tamil. The catalog includes pricing data, discount percentages, stock quantities, and bestseller flags.

| Metric                   | Value                              |
|--------------------------|------------------------------------|
| Total Books              | 25                                 |
| Bestsellers              | 17 (68%)                           |
| Non-Bestsellers          | 8 (32%)                            |
| Average Sale Price (INR) | 445.24                             |
| Price Range (INR)        | 254.15 – 809.10                    |
| Languages Available      | English (15), Hindi (7), Tamil (3) |

### 3.2 Category Distribution

| Category   | No. of Books | % of Catalog |
|------------|--------------|--------------|
| Self-Help  | 7            | 28%          |
| Fiction    | 6            | 24%          |
| Business   | 4            | 16%          |
| Technology | 4            | 16%          |
| Education  | 2            | 8%           |
| Biography  | 1            | 4%           |
| History    | 1            | 4%           |

### 3.3 Top Rated Books

| Book Title                          | Category  | Avg Rating | Bestseller |
|-------------------------------------|-----------|------------|------------|
| The 5 AM Club                       | Self-Help | 5.0        | Yes        |
| Think and Grow Rich                 | Business  | 4.9        | Yes        |
| The Subtle Art of Not Giving a F*ck | Self-Help | 4.9        | Yes        |
| Atomic Habits                       | Self-Help | 4.8        | No         |
| God of Small Things                 | Fiction   | 4.8        | Yes        |
| Deep Work                           | Self-Help | 4.7        | Yes        |
| Two States                          | Fiction   | 4.6        | Yes        |

### 3.4 Complete Book Catalog

| Book ID | Title                                 | Category   | Sale Price (INR) | Rating | Bestseller |
|---------|---------------------------------------|------------|------------------|--------|------------|
| B101    | Harry Potter and the Sorcerer's Stone | Fiction    | 474.05           | 3.9    | Yes        |
| B102    | The Alchemist                         | Self-Help  | 254.15           | 3.5    | No         |
| B103    | Atomic Habits                         | Self-Help  | 569.05           | 4.8    | No         |
| B104    | Rich Dad Poor Dad                     | Business   | 427.50           | 3.7    | Yes        |
| B105    | Python Crash Course                   | Technology | 594.15           | 3.7    | No         |
| B106    | Data Analytics Guide                  | Education  | 756.90           | 3.6    | Yes        |
| B107    | SQL for Beginners                     | Technology | 585.00           | 4.3    | No         |

| Book ID | Title                               | Category   | Sale Price (INR) | Rating | Bestseller |
|---------|-------------------------------------|------------|------------------|--------|------------|
| B108    | Machine Learning Essentials         | Technology | 809.10           | 3.9    | Yes        |
| B109    | Wings of Fire                       | Biography  | 560.00           | 3.6    | Yes        |
| B110    | The Secret                          | Self-Help  | 326.05           | 3.8    | Yes        |
| B111    | Deep Work                           | Self-Help  | 339.15           | 4.7    | Yes        |
| B112    | The Psychology of Money             | Business   | 424.15           | 4.5    | No         |
| B113    | Sapiens                             | History    | 299.05           | 3.6    | No         |
| B114    | Zero to One                         | Business   | 509.05           | 3.9    | Yes        |
| B115    | The 5 AM Club                       | Self-Help  | 424.15           | 5.0    | Yes        |
| B116    | Ikigai                              | Self-Help  | 339.15           | 3.9    | No         |
| B117    | Think and Grow Rich                 | Business   | 409.05           | 4.9    | Yes        |
| B118    | Data Structures in C++              | Technology | 424.15           | 3.9    | Yes        |
| B119    | God of Small Things                 | Fiction    | 474.05           | 4.8    | Yes        |
| B120    | The White Tiger                     | Fiction    | 319.05           | 3.7    | No         |
| B121    | Train to Pakistan                   | Fiction    | 354.15           | 4.1    | Yes        |
| B122    | Two States                          | Fiction    | 339.15           | 4.6    | Yes        |
| B123    | The Immortals of Meluha             | Fiction    | 449.05           | 3.9    | Yes        |
| B124    | The Subtle Art of Not Giving a F*ck | Self-Help  | 569.05           | 4.9    | Yes        |
| B125    | Digital Marketing Masterclass       | Education  | 509.05           | 4.5    | Yes        |

## 4. Customer Analysis

### 4.1 Customer Demographics

The platform serves 60 registered customers from various cities across India. The customer base is evenly split by gender and spans a working-age demographic.

| Metric          | Value                             |
|-----------------|-----------------------------------|
| Total Customers | 60                                |
| Gender Split    | Male: 30 (50%)   Female: 30 (50%) |
| Age Range       | 19 – 45 years                     |
| Average Age     | 31.4 years                        |

## 4.2 Membership Distribution

| Membership Tier | No. of Customers | Percentage |
|-----------------|------------------|------------|
| Free            | 20               | 33.3%      |
| Silver          | 15               | 25.0%      |
| Gold            | 14               | 23.3%      |
| Platinum        | 11               | 18.3%      |

One-third of the customer base is on the free tier, indicating strong growth potential for membership conversion campaigns targeting Silver, Gold, and Platinum upgrades.

## 4.3 Geographic Distribution (Top Cities)

| City          | Customers | State          |
|---------------|-----------|----------------|
| Coimbatore    | 6         | Tamil Nadu     |
| Vijayawada    | 5         | Andhra Pradesh |
| Chandigarh    | 5         | Punjab         |
| Visakhapatnam | 5         | Andhra Pradesh |
| Indore        | 4         | Madhya Pradesh |

## 4.4 Preferred Book Categories

| Category   | Customers Preferring |
|------------|----------------------|
| Education  | 12                   |
| Biography  | 12                   |
| History    | 10                   |
| Self-Help  | 8                    |
| Business   | 8                    |
| Technology | 5                    |
| Fiction    | 5                    |

## 5. Orders & Revenue Analysis

### 5.1 Overview

The platform recorded 150 orders generating total revenue of INR 1,76,689.30 between January and June 2025. The average order value stands at INR 1,177.93.

### 5.2 Monthly Revenue Trend

| Month         | Revenue (INR) | Growth   |
|---------------|---------------|----------|
| January 2025  | 26,692.80     | Baseline |
| February 2025 | 30,077.20     | +12.7%   |
| March 2025    | 33,131.55     | +10.2%   |
| April 2025    | 34,624.40     | +4.5%    |
| May 2025      | 24,861.55     | -28.2%   |
| June 2025     | 27,301.80     | +9.8%    |

Revenue grew steadily from January to April 2025, peaking at INR 34,624. A notable dip in May 2025 may reflect seasonal factors or a reduction in marketing activity.

### 5.3 Order Status Distribution

| Order Status | Count | Percentage |
|--------------|-------|------------|
| Delivered    | 93    | 62.0%      |
| Shipped      | 16    | 10.7%      |
| Processing   | 14    | 9.3%       |
| Cancelled    | 13    | 8.7%       |
| Returned     | 9     | 6.0%       |
| Pending      | 5     | 3.3%       |

### 5.4 Payment Methods

| Payment Method | Orders | Share |
|----------------|--------|-------|
| Debit Card     | 31     | 20.7% |
| Credit Card    | 28     | 18.7% |
| UPI            | 25     | 16.7% |

| Payment Method   | Orders | Share |
|------------------|--------|-------|
| Cash on Delivery | 25     | 16.7% |
| Net Banking      | 21     | 14.0% |
| Wallet           | 20     | 13.3% |

5.5 Payment Success Rate

| Payment Status | Count | Percentage |
|----------------|-------|------------|
| Success        | 132   | 88.0%      |
| Failed         | 13    | 8.7%       |
| Pending        | 5     | 3.3%       |

5.6 Coupon Usage Analysis

| Coupon Code | Orders Used | Share |
|-------------|-------------|-------|
| No Coupon   | 69          | 46.0% |
| WELCOME20   | 28          | 18.7% |
| SUMMER15    | 27          | 18.0% |
| SAVE10      | 26          | 17.3% |

5.7 Top 5 Best-Selling Books (by Units)

| Rank | Book Title                            | Category   | Units Sold |
|------|---------------------------------------|------------|------------|
| 1    | Harry Potter and the Sorcerer's Stone | Fiction    | 27         |
| 2    | SQL for Beginners                     | Technology | 27         |
| 3    | Think and Grow Rich                   | Business   | 21         |
| 4    | Data Structures in C++                | Technology | 21         |
| 5    | Rich Dad Poor Dad                     | Business   | 20         |



## 6. Returns & Refund Analysis

### 6.1 Returns Overview

A total of 9 returns were processed against 150 orders, yielding a return rate of 6.0%. The total refund value amounts to INR 13,114.10 with an average return window of 7.9 days.

| Metric                   | Value     |
|--------------------------|-----------|
| Total Returns            | 9         |
| Return Rate              | 6.0%      |
| Total Refund Value (INR) | 13,114.10 |
| Average Days to Return   | 7.9 days  |

### 6.2 Return Reasons

| Return Reason       | Count | Percentage |
|---------------------|-------|------------|
| Changed my mind     | 3     | 33.3%      |
| Not as described    | 2     | 22.2%      |
| Accidental purchase | 2     | 22.2%      |
| Damaged on arrival  | 1     | 11.1%      |
| Duplicate order     | 1     | 11.1%      |

### 6.3 Refund Status

| Refund Status | Count | Percentage |
|---------------|-------|------------|
| Rejected      | 5     | 55.6%      |
| Processed     | 3     | 33.3%      |
| Pending       | 1     | 11.1%      |

A notably high rejection rate (55.6%) for refunds suggests that refund policy criteria may be strictly applied. Reviewing rejection criteria could improve customer satisfaction and reduce dispute escalations.

## 6.4 Complete Returns Log

| Return ID | Book Title          | Return Reason       | Refund (INR) | Status    |
|-----------|---------------------|---------------------|--------------|-----------|
| RET0001   | Rich Dad Poor Dad   | Changed my mind     | 427.50       | Processed |
| RET0002   | Wings of Fire       | Changed my mind     | 560.00       | Rejected  |
| RET0003   | The Alchemist       | Accidental purchase | 1,196.00     | Rejected  |
| RET0004   | Sapiens             | Not as described    | 897.15       | Processed |
| RET0005   | Ikigai              | Damaged on arrival  | 678.30       | Pending   |
| RET0006   | Harry Potter        | Duplicate order     | 1,796.40     | Processed |
| RET0007   | The Secret          | Not as described    | 978.15       | Rejected  |
| RET0008   | Atomic Habits       | Accidental purchase | 2,277.00     | Rejected  |
| RET0009   | Python Crash Course | Changed my mind     | 3,303.60     | Rejected  |

## 7. Customer Reviews & Ratings Analysis

### 7.1 Reviews Overview

The platform has received 200 customer reviews across all book titles. The overall average rating is 3.78 out of 5.0, with 73% of reviews coming from verified purchasers. The average helpful vote count per review is 71.8, indicating strong community engagement.

| Metric                           | Value       |
|----------------------------------|-------------|
| Total Reviews                    | 200         |
| Average Rating                   | 3.78 / 5.0  |
| Verified Purchase Reviews        | 146 (73.0%) |
| Unverified Reviews               | 54 (27.0%)  |
| Average Helpful Votes per Review | 71.8        |

### 7.2 Rating Distribution

| Star Rating | Number of Reviews | Percentage |
|-------------|-------------------|------------|
| 5 Stars     | 70                | 35.0%      |
| 4 Stars     | 66                | 33.0%      |
| 3 Stars     | 29                | 14.5%      |

| Star Rating | Number of Reviews | Percentage |
|-------------|-------------------|------------|
| 2 Stars     | 20                | 10.0%      |
| 1 Star      | 15                | 7.5%       |

A majority of reviews (68%) are positive (4 or 5 stars), indicating overall customer satisfaction with the product offerings.

7.3 Most Reviewed Books

| Book Title          | Review Count | Avg Rating |
|---------------------|--------------|------------|
| Zero to One         | 14           | 4.00       |
| The Secret          | 13           | 2.62       |
| Sapiens             | 11           | 3.55       |
| Train to Pakistan   | 11           | 4.09       |
| Think and Grow Rich | 10           | 4.30       |

8. Key Business Insights & Recommendations

8.1 Strengths

- Strong delivery rate of 62% with consistent monthly revenue growth (Jan-Apr 2025)
- High bestseller ratio (68%) signals a well-curated, market-aligned catalog
- Diverse payment ecosystem covering 6 methods, reducing checkout friction
- 73% verified purchase rate on reviews lends credibility to ratings data
- Balanced gender distribution (50/50) enables unbiased marketing strategies

8.2 Opportunities

- 33.3% of customers are on the Free tier — targeted upgrade campaigns (Silver/Gold) could drive subscription revenue
- WELCOME20 and SUMMER15 coupons each drove ~18-19% of orders — seasonal coupon strategies are effective and should be expanded
- Education and Biography are the most preferred categories but are underrepresented in the catalog (3 titles total) — stocking more titles here is a priority
- Andhra Pradesh and Tamil Nadu dominate the top-city list — regional language cataloging (Telugu) could boost engagement

### 8.3 Areas for Improvement

- May 2025 revenue dip of 28.2% warrants investigation — seasonal analysis or marketing gap review recommended
- 8.7% payment failure rate should be investigated to identify technical or UX issues at checkout
- 55.6% refund rejection rate may indicate overly strict refund policy — consider revising return conditions to improve customer trust
- The 6% return rate, while moderate, is driven by avoidable reasons (changed mind, accidental purchase) — better product descriptions and purchase confirmation flows could reduce these

### 8.4 Data Quality Notes

- Returns dataset has only 9 records — further data collection recommended for statistically significant return trend analysis
- Review count per book varies widely (1-14) — consider prompting reviews post-purchase to normalize coverage
- Customer registration dates range from 2023 onward — longitudinal customer lifetime value (LTV) analysis is now feasible

## 9. Technical Appendix

### 9.1 Field Descriptions — Books.csv

| Field            | Type    | Description                                          |
|------------------|---------|------------------------------------------------------|
| Book_Id          | String  | Unique identifier for each book (e.g., B101)         |
| Book_Title       | String  | Title of the book                                    |
| Author           | String  | Author name                                          |
| Category         | String  | Genre/category (Fiction, Self-Help, Business, etc.)  |
| Publication_Year | Integer | Year the book was published                          |
| Publisher        | String  | Publishing company                                   |
| Language         | String  | Book language (English, Hindi, Tamil)                |
| Pages            | Integer | Number of pages                                      |
| Unit_Price_Inr   | Float   | Original price in Indian Rupees                      |
| Discount_Pct     | Float   | Discount percentage applied                          |
| Sale_Price_Inr   | Float   | Discounted sale price in INR                         |
| Stock_Quantity   | Integer | Units available in stock                             |
| Avg_Rating       | Float   | Average rating (out of 5)                            |
| Num_Reviews      | Integer | Total number of reviews received                     |
| Is_Bestseller    | String  | Whether the book is flagged as a bestseller (Yes/No) |

9.2 Field Descriptions — customers.csv

| Field              | Type    | Description                                    |
|--------------------|---------|------------------------------------------------|
| Customer_Id        | String  | Unique identifier (e.g., C001)                 |
| Customer_Name      | String  | Full name of the customer                      |
| Gender             | String  | Male or Female                                 |
| Age                | Integer | Age of the customer                            |
| Email              | String  | Contact email address                          |
| Phone Number       | String  | Contact phone number                           |
| City               | String  | City of residence                              |
| State              | String  | State of residence                             |
| Registration_Date  | Date    | Date of account registration                   |
| Membership         | String  | Membership tier (Free, Silver, Gold, Platinum) |
| Preferred_Category | String  | Customer's preferred book category             |

9.3 Field Descriptions — Orders.csv

| Field            | Type    | Description                                                       |
|------------------|---------|-------------------------------------------------------------------|
| Order_Id         | Integer | Unique order identifier                                           |
| Order_Date       | Date    | Date the order was placed                                         |
| Customer_Id      | String  | FK reference to customers.csv                                     |
| Book_Id          | String  | FK reference to Books.csv                                         |
| Quantity_Ordered | Integer | Number of copies ordered                                          |
| Total_Amount_Inr | Float   | Total order value in INR                                          |
| Payment_Method   | String  | Payment method used                                               |
| Payment_Status   | String  | Success / Failed / Pending                                        |
| Order_Status     | String  | Delivered / Shipped / Processing / Cancelled / Returned / Pending |
| Coupon_Used      | String  | Coupon code applied or 'No Coupon'                                |

9.4 Field Descriptions — Returns.csv

| Field         | Type    | Description                              |
|---------------|---------|------------------------------------------|
| Return_Id     | String  | Unique return identifier (e.g., RET0001) |
| Order_Id      | Integer | FK reference to Orders.csv               |
| Customer_Id   | String  | FK reference to customers.csv            |
| Customer_Name | String  | Customer's full name                     |

| Field               | Type    | Description                                       |
|---------------------|---------|---------------------------------------------------|
| Book_Id             | String  | FK reference to Books.csv                         |
| Book_Title          | String  | Title of the returned book                        |
| Category            | String  | Book category                                     |
| Purchase_Date       | Date    | Original purchase date                            |
| Return_Date         | Date    | Date return was initiated                         |
| Days_To_Return      | Integer | Days between purchase and return                  |
| Return_Reason       | String  | Reason stated by customer                         |
| Original_Amount_Inr | Float   | Original purchase price                           |
| Refund_Amount_Inr   | Float   | Refund amount issued                              |
| Refund_Method       | String  | Store Credit / Original Payment                   |
| Refund_Status       | String  | Processed / Rejected / Pending                    |
| Return_Condition    | String  | Condition of returned item (Acceptable / Damaged) |

## 9.5 Field Descriptions — Reviews\_1.csv

| Field             | Type    | Description                                      |
|-------------------|---------|--------------------------------------------------|
| Review_Id         | String  | Unique review identifier (e.g., REV0001)         |
| Book_Id           | String  | FK reference to Books.csv                        |
| Book_Title        | String  | Title of the reviewed book                       |
| Category          | String  | Book category                                    |
| Customer_Id       | String  | FK reference to customers.csv                    |
| Customer_Name     | String  | Reviewer's name                                  |
| Rating            | Integer | Star rating (1–5)                                |
| Review_Text       | String  | Free-text review content                         |
| Review_Date       | Date    | Date review was submitted                        |
| Helpful_Votes     | Integer | Number of users who found the review helpful     |
| Verified_Purchase | String  | Whether the reviewer purchased the book (Yes/No) |